

CHANCE

The laws of luck

Here's a handy tip. In maths questions about probability, look out for the words "or" and "and". When you see the word "or", chances are you'll need to **add up** probabilities to get your answer. So the chance of rolling a one *or* a two with a die would be $\frac{1}{6} + \frac{1}{6} = \frac{2}{6}$. When you see "and", you'll probably have to **multiply**. For instance, the chance of getting a six *and* another six on two dice rolls is $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

Luck of the draw

If you shuffle a pack of cards, what's the probability that the cards will end up in one particular order? The answer is 1 over the total number of card combinations possible.

We can work out the total number of card combinations like this:

$$52 \times 51 \times 50 \times 49 \times \dots \times 1$$

(We write the sum in shorthand as 52!)

This sum produces a **very** big number:

80 *million trillion trillion trillion trillion trillion*. So the chance that your cards are in one particular order is 1 in 80 million

trillion trillion trillion

trillion trillion. Next time

you shuffle a pack of cards, think

about this: it's very likely that nobody

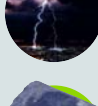
in the history of the universe has ever had

exactly the same order of cards before!



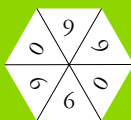
RISKY BUSINESS

Some people are terrified of lightning but happy to smoke. If they understood probability, they might think differently. The table shows your chance of dying from various causes, based on death rates in Europe and North America.

CAUSE OF DEATH	CHANCE OF DYING IN A YEAR
 <i>smoking 10 cigarettes a day</i>	1 in 200
 <i>heart attack</i>	1 in 300
 <i>road accident</i>	1 in 4000
 <i>flu</i>	1 in 5000
 <i>falling</i>	1 in 16,000
 <i>playing football</i>	1 in 25,000
 <i>murder</i>	1 in 100,000
 <i>struck by lightning</i>	1 in 10 million
 <i>hit by meteorite (estimate)</i>	1 in a trillion

SNEAKY SPINNERS

Here's a game of chance you'll keep winning at. Make four spinners like the ones here by cutting out hexagons of card and writing numbers on them. Push a cocktail stick through the centre of each. Then challenge a friend to a spinner match. Point out that they can choose any spinner they want, and



the numbers on each one add up to 24, so the game must be fair. Look at the highest number on their spinner, then make sure you pick the spinner with the next number up (but if your friend takes the one with an 8 on it, you take the one with a 5 on it). You'll have a two-thirds chance of winning each match!

